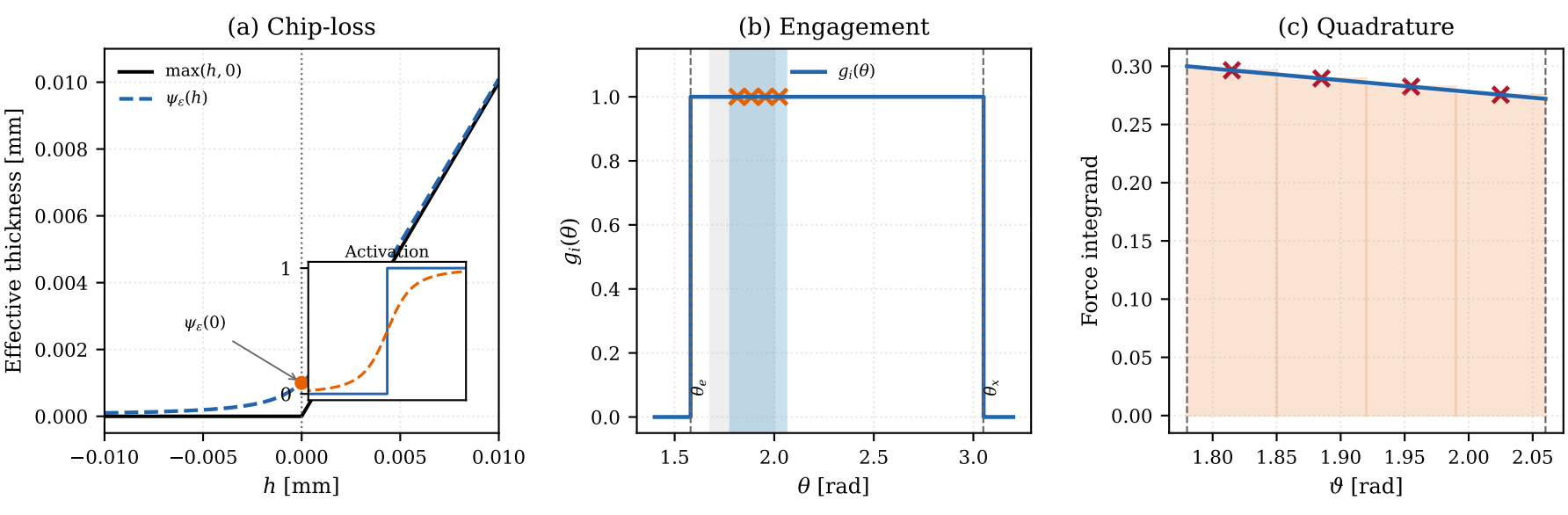
(a) Chip-loss
max(h, 0)
ψε(h)
Activation
1
ψε(0)
θ*
Effective thickness [mm]
h [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.010
−0.005
0.000
0.005
0.010
(b) Engagement
gi(θ)
θe
θx
θ [rad]
1.0
0.8
0.6
0.4
0.2
0.0
1.5
2.0
2.5
3.0
(c) Quadrature
Force integrand
ϑ [rad]
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1.80
1.85
1.90
1.95
2.00
2.05